

Course Syllabus

010153002 Computer Programming

Semester 1, Academic Year 2026 — Faculty of Engineering, King Mongkut's University of Technology North Bangkok

Revised May 27, 2026

Course Information

Course Code	010153002
Course Title	Computer Programming
Credit Hours	3 credits (2 lecture hours + 2 laboratory hours per week)
Semester	1 / 2026
Schedule	Every Thursday, 13:00–17:00
Lecture	13:00–15:00 (2 hours)
Laboratory	15:00–17:00 (2 hours)
Semester Dates	22 June 2026 – 11 October 2026
First Class	Thursday, 25 June 2026
Last Class	Thursday, 24 September 2026
Instructor	Dr. Ruslee Sutthaweekul
Language	English (supplementary explanations in Thai available)
Prerequisites	None

Course Description

An introduction to computer programming using the C language. Topics cover the complete programming workflow: writing, compiling, and debugging programs; primitive data types; operators and expressions; control flow (selection and iteration); functions; arrays; pointers and dynamic memory; strings; structures, unions, and enumerations; and file input/output. Students apply each concept during weekly two-hour laboratory sessions using hands-on programming exercises.

Learning Outcomes

Upon successful completion of this course, students will be able to:

- LO1.** Write, compile, and run syntactically correct C programs from scratch.
- LO2.** Apply appropriate data types, operators, and control structures to solve computational problems.
- LO3.** Design, implement, and test functions with well-defined parameters, return types, and scope.
- LO4.** Use arrays and pointers to manage sequences of data and dynamic memory.
- LO5.** Manipulate strings using the standard `string.h` library and pointer-based traversal.
- LO6.** Define and use `struct`, `union`, and `enum` types to model structured data.
- LO7.** Perform file input/output using the standard C file API (`FILE *`, `fopen`, `fscanf`, `fprintf`).
- LO8.** Collaborate effectively with AI coding assistants while maintaining critical evaluation of generated code.

Required Materials

Item	Details
Lecture slides	Available on the course website before each class
Lab sheets	Distributed as PDF each week; bring a printed copy
Compiler	GCC (Linux/macOS) or MinGW-GCC (Windows); any IDE is fine
Source code	Provided in the course repository (<code>src/lec01–lec12</code>)
<i>Recommended</i>	<i>Kernighan & Ritchie, “The C Programming Language,” 2nd ed.</i>

Assessment & Grading

Component	Description	Weight
Attendance	Present and engaged; $\geq 80\%$ required to sit the final exam	10 %
Lab Exercises	10 in-lab worksheets (Labs 1–10), graded on completion and correctness	20 %
Project Assignment	One individual programming project (announced Week 9)	10 %
Midterm Examination	Covers Part 1 (Lectures 1–7); date and venue TBD by Registrar	30 %
Final Examination	Covers Part 2 only (Lectures 8–10); date and venue TBD by Registrar	30 %
Total		100 %

Grading Scale

A $\geq 90\%$ **B+** 85–89 % **B** 80–84 % **C+** 75–79 % **C** 70–74 % **D+** 65–69 %
D 60–64 % **F** $< 60\%$

Examination Policy

- Both examinations are **closed-book and paper-based** (no computer, no AI tools).
- Students must bring their student ID card to enter the exam room.
- Questions include multiple-choice, code tracing, bug fixing, and short programming tasks (written by hand).
- No make-up exams except with documented medical or official university reasons, approved *before* the exam date wherever possible.
- A minimum attendance of 80 % is required to be eligible to sit the final exam.

Academic Integrity & AI Collaboration Policy

Using AI Tools in This Course

AI assistants (ChatGPT, Claude, Copilot, ...) can complete these lab exercises in seconds. You must decide each time what you actually want to learn.

Permitted use:

- Ask the AI to *explain* a concept or error message in a different way.
- Ask for *hints* rather than solutions; use the AI as a tutor, not an answer machine.
- Use AI to review your own code *after* you have written it.

Not permitted:

- Submitting AI-generated code as your own lab or project work.
- Copying a solution you cannot reproduce yourself, by hand, five minutes later.

The litmus test: If asked to write the same program on paper in the exam, could you? If yes, your AI use was productive. If no, you are cheating yourself.

Plagiarism and academic dishonesty are subject to university disciplinary procedures, which may include a grade of F for the course.

Weekly Schedule

All classes meet on **Thursday, 13:00–17:00**. Lecture: 13:00–15:00 — Laboratory: 15:00–17:00.

Week	Date	Lecture (13:00–15:00)	Lab (15:00–17:00)
1	Thu 25 Jun 2026	Lec 1: Introduction & Overview Course structure, compilation pipeline, first program, development environment.	Lab 1: Introduction to C Compile and run <code>hello_world.c</code> ; explore the compiler flags.
2	Thu 02 Jul 2026	Lec 2: Basic Data Types & Expressions Integers, floats, chars; <code>printf/scanf</code> ; arithmetic & assignment operators.	Lab 2: Expressions & Operators Type conversion, pre/post increment, operator precedence.
3	Thu 09 Jul 2026	Lec 3: Sequential & Conditional Flow <code>if</code> , <code>if-else</code> , nested <code>if</code> , <code>switch</code> ; ternary operator.	Lab 3: Conditional Statements Grade classifier, switch on user input.
4	Thu 16 Jul 2026	Lec 4: Loop Control Structures <code>while</code> , <code>do-while</code> , <code>for</code> ; <code>break</code> , <code>continue</code> ; debugging with <code>printf</code> .	Lab 4: Loop Control Summation, factorial, pattern printing, input validation.
5	Thu 23 Jul 2026	Lec 5: Functions & Macros Declaration, definition, call, return values; scope; recursion; <code>#define</code> macros.	Lab 5: Functions Write and call helper functions; implement recursive factorial.
6	Thu 30 Jul 2026	Lec 6: Arrays 1-D and 2-D arrays; traversal; passing arrays to functions; <code>sizeof</code> decay.	Lab 6: Arrays Statistics on scores array; bubble sort; matrix addition.
7	Thu 06 Aug 2026	Lec 7: Pointers Address-of & dereference; pointer arithmetic; <code>malloc/free</code> ; function pointers.	Lab 7: Pointers Pointer tracing, swap via pointers, dynamic array allocation.

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Week	Date	Lecture (13:00–15:00)	Lab (15:00–17:00)
8	Thu 13 Aug 2026	Part 1 review & consolidation Q&A on Lectures 1–7; past exam question walkthrough; common pitfalls.	Lab 8: Strings <code>my_strlen</code> ; reverse in place; palindrome checker. (<i>Lec 8 content distributed as pre-reading.</i>)
—	Thu 20 Aug 2026	University Midterm Examination Period (17–23 August 2026) No class this week. Midterm exam date & venue TBD by the Registrar. Covers: Lectures 1–7, Labs 1–7.	
9	Thu 27 Aug 2026	Lec 8: Strings (with Pointers) Null-terminated <code>char</code> arrays; <code>strlen</code> , <code>strcpy</code> , <code>strcmp</code> , <code>strcat</code> ; pointer traversal.	Lab 8: Strings (continued) Count vowels; palindrome with two pointers.
10	Thu 03 Sep 2026	Lec 9: Structures, Unions & Enumerations <code>struct</code> definition; <code>.</code> vs <code>-></code> ; pass-by-pointer; <code>typedef</code> ; <code>union</code> ; <code>enum</code> ; memory padding.	Lab 9: Structures & Unions Student record with <code>-></code> ; sort structs; tagged union area calculator. <i>Project assignment announced.</i>
11	Thu 10 Sep 2026	Lec 10: File Operations <code>FILE *</code> ; <code>fopen/fclose</code> ; <code>fscanf/fgets/fgetc</code> ; <code>fprintf</code> ; modes; EOF detection.	Lab 10: File Operations Copy a file with <code>fgets</code> ; sum integers from file; student report sorted by score.
12	Thu 17 Sep 2026	Project work session Instructor and TA available for project Q&A; individual code review on request.	Course review & final exam preparation Past exam questions; common bugs; tracing exercises.
13	Thu 24 Sep 2026	Course wrap-up & Q&A Open Q&A on all topics; exam strategy; feedback session. Last class. <i>Project submission deadline: 17:00 today.</i>	
—	TBD	Final Examination <i>Date & venue TBD by the Registrar</i> Covers Part 2 only : Lectures 8–10 (Strings, Structs, File Operations) and Labs 8–10. Closed-book; no computer; no AI tools.	

Note: Teaching concludes on **24 September 2026** (Week 13). The university midterm exam period (17–23 Aug) and final exam dates are set by the Registrar and will be announced on the LMS as soon as confirmed. The **final exam covers Part 2 only** (Lectures 8–12). Students requiring accommodation must contact the instructor before the last class.

Course Policies

Attendance Students must attend at least 80% of class sessions (12 of 14) to be eligible to sit the Final Examination. Notify the instructor *before* class if you must be absent.

Lab Work Lab sheets must be submitted by the end of the lab session. Late submissions are not accepted without a valid excuse approved in advance. Students are expected to type and run their own code during the lab.

Project The project is an individual assignment; collaboration on code is not permitted. Submitted code must compile and run on GCC without modification. Include a short written

report (one page) describing your design choices.

Communication The primary communication channel is the course LMS (announced in Week 1). Email responses can be expected within two working days.

Devices Laptops and tablets are permitted during lecture and lab for course-related work only. Mobile phones should be silenced.

Topic–Outcome Alignment

Lecture	Learning Outcomes Addressed
Lec 1 – Introduction	LO1
Lec 2 – Data Types	LO1, LO2
Lec 3 – Conditionals	LO2
Lec 4 – Loops	LO2
Lec 5 – Functions	LO3
Lec 6 – Arrays	LO4
Lec 7 – Pointers	LO4
Lec 8 – Strings	LO5
Lec 9 – Structs & Unions	LO6
Lec 10 – File Operations	LO7
All lectures	LO8 (AI collaboration)

Questions about this syllabus? Contact Dr. Ruslee Sutthaweekul.

This document is subject to revision; changes will be announced in class and on the LMS.